

ASSIGNMENT #06

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
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Due Date: Monday, November 10th in class (11:00AM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

CHE301 HW6

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Problem A

We want to prove the following thermodynamic identity. We will do this by consulting ChatGPT and confirming our answers.

$$\int_0^P (1 - \beta T) V dP = RT(Z - 1) - \int_0^P RT^2 \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho} \quad \text{where} \quad Z = \frac{PV}{RT}$$

First, it will be helpful to define what each variable represents. Reference the table below.

Variable	Parameter
P	Pressure
β	Thermal Expansion Coefficient
T	Temperature
V	Molar Volume
R	Gas Constant
Z	Compressibility Factor
ρ	Molar Density

Now, let's define the relationship between a few variables. The molar density is the inverse of the molar volume. The compressibility factor is defined below.

$$\rho = \frac{1}{V} \qquad Z = \frac{PV}{RT} = \frac{P}{\rho RT}$$

To begin our derivation, let's first recognize the definition of the thermal expansion coefficient that we will be using. The definition below is from the textbook.

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_\rho$$

Multiply both sides by T and V to yield the following.

$$\beta TV = T \left(\frac{\partial V}{\partial T} \right)_\rho$$

Multiply both sides by -1 and add V to yield the following.

$$V - \beta TV = V - T \left(\frac{\partial V}{\partial T} \right)_P$$

Now, factor out V . We designate this equation as equation 1 so we can easily reference it later.

$$(1 - \beta T)V = V - T \left(\frac{\partial V}{\partial T} \right)_\rho \quad (1)$$

To continue our derivation, let's define the left hand side and right hand sides of the thermodynamic identity as separate functions of pressure and temperature. Let's call the left hand side $X(P, T)$ and the right hand side $Y(P, T)$. The goal of this proof is to prove that the differentials of both sides are equal to one another, and then to show that both sides are equal to zero when the pressure is at zero.

$$X(P, T) = \int_0^P (1 - \beta T)V dP$$

$$Y(P, T) = RT(Z - 1) - \int_0^\rho RT^2 \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho}$$

Continue by differentiating $X(P, T)$ with respect to pressure at constant temperature to yield the following.

$$\left(\frac{\partial X}{\partial P} \right)_T = (1 - \beta T)V$$

Recognize that the right hand side of the above equation is identical to the left hand side of equation 1. So, we substitute that in and call this equation 2.

$$\left(\frac{\partial X}{\partial P} \right)_T = V - T \left(\frac{\partial V}{\partial T} \right)_\rho \quad (2)$$

Now, let's differentiate $Y(P, T)$. We start by taking the differential of the first term with respect to pressure at constant temperature. Recognizing that R and T are constants yields the following.

$$\frac{\partial}{\partial P}[RT(Z - 1)] = RT \left(\frac{\partial Z}{\partial P} \right)_T$$

Now, substitute $\frac{PV}{RT}$ in for Z to get the following

$$\frac{\partial}{\partial P}[RT(Z - 1)] = RT \left[\frac{\partial}{\partial P} \left(\frac{PV}{RT} \right) \right]_T$$

Now expand this term out by differentiating and using the product rule to yield the following

$$\frac{\partial}{\partial P}[RT(Z - 1)] = RT \left(\frac{V + P \left(\frac{\partial V}{\partial P} \right)_T}{RT} \right)$$

Now cancel out the RT from the numerator and denominator to yield the following.

$$\frac{\partial}{\partial P}[RT(Z - 1)] = V + P \left(\frac{\partial V}{\partial P} \right)_T$$

Next, let's differentiate the integrand term in $Y(P, T)$. To do this, we have to recognize a useful calculus identity known as Leibniz's rule below. It states that differentiating an integral with an upper limit and integrand both dependent on a value x results in the following.

$$\frac{d}{dx} \int_a^{b(x)} f(y, x) dy = f(b(x), x) \frac{db}{dx} + \int_a^{b(x)} \frac{\partial f}{\partial x} dy$$

Applying this to the integrand term in $Y(P, T)$ yields the following.

$$f(\rho, T) = RT^2 \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{1}{\rho}$$

$$-\frac{\partial}{\partial P} \int_0^{\rho(P,T)} f(\rho, T) d\rho = -f(\rho, T) \frac{\partial \rho}{\partial P} - \int_0^{\rho} \frac{\partial f}{\partial P} d\rho$$

To clear potential ambiguity, let's state here that the compressibility factor is a function solely of molar density and temperature. In other words, $Z = Z(\rho, T)$. Knowing this, we can recognize that the remaining integrand term after using the Leibniz rule is equal to zero, since our function f will not change with pressure. After removing the integrand term and substituting f back in, the following equation is obtained.

$$-\frac{\partial}{\partial P} \left[\int_0^{\rho} RT^2 \left(\frac{\partial Z}{\partial T} \right)_{\rho} \frac{d\rho}{\rho} \right]_T = -RT^2 \left(\frac{\partial Z}{\partial T} \right)_{\rho} \frac{1}{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T$$

Now we can combine both terms for $Y(P, T)$ to get equation 3.

$$\left(\frac{\partial Y}{\partial P} \right)_T = V + P \left(\frac{\partial V}{\partial P} \right)_T - RT^2 \left(\frac{\partial Z}{\partial T} \right)_{\rho} \frac{1}{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T \quad (3)$$

Now, we will perform algebraic manipulations on equation 3 so that it is equal to equation 2. Our first step in this is to re-express $\left(\frac{\partial Z}{\partial T} \right)_{\rho}$ using the relationship $Z = \frac{P}{\rho RT}$.

$$\left(\frac{\partial Z}{\partial T} \right)_{\rho} = \frac{\partial}{\partial T} \left(\frac{P}{\rho RT} \right)$$

Recognize P is a function of T , so we must differentiate using product rule.

$$\left(\frac{\partial Z}{\partial T} \right)_{\rho} = \frac{1}{\rho R} \frac{\partial}{\partial T} \left(\frac{P}{T} \right)_{\rho} = \frac{1}{\rho R} \left[\frac{1}{T} \left(\frac{\partial P}{\partial T} \right)_{\rho} - \frac{P}{T^2} \right]$$

Multiply by RT^2 and again use $Z = \frac{P}{\rho RT}$ to get the following. Let's call this equation 4.

$$RT^2 \left(\frac{\partial Z}{\partial T} \right)_{\rho} = \frac{T}{\rho} \left(\frac{\partial P}{\partial T} \right)_{\rho} - \frac{P}{\rho} \quad (4)$$

Recognize $V = \frac{1}{\rho}$. Use this relation, and differentiate using chain rule to get the following. Call this equation 5

$$\left(\frac{\partial V}{\partial P} \right)_T = -\frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial P} \right)_T \quad (5)$$

Now input equations 4 and 5 into equation 3 to get the following.

$$\begin{aligned} \left(\frac{\partial Y}{\partial P} \right)_T &= V - \frac{P}{\rho^2} \left(\frac{\partial \rho}{\partial P} \right)_T - \frac{T}{\rho^2} \left(\frac{\partial P}{\partial T} \right)_{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T + \frac{P}{\rho^2} \left(\frac{\partial \rho}{\partial P} \right)_T \\ \left(\frac{\partial Y}{\partial P} \right)_T &= V - \frac{T}{\rho^2} \left(\frac{\partial P}{\partial T} \right)_{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T \end{aligned}$$

Now, using a similar differentiation with chain rule that we used to obtain equation 5 gives us the following relation.

$$\left(\frac{\partial V}{\partial T} \right)_P = -\frac{1}{\rho^2} \left(\frac{\partial \rho}{\partial T} \right)_P$$

Now we add and subtract the $T \left(\frac{\partial V}{\partial T} \right)_P$ while using the above relation to get the following. Call this equation 6.

$$\left(\frac{\partial Y}{\partial P} \right)_T = V - T \left(\frac{\partial V}{\partial T} \right)_P + \frac{T}{\rho^2} \left[-\left(\frac{\partial P}{\partial T} \right)_{\rho} \left(\frac{\partial \rho}{\partial P} \right)_T - \left(\frac{\partial \rho}{\partial T} \right)_P \right] \quad (6)$$

To proceed, we must utilize the cyclic identity that was taught in class. It is given below.

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$$

And adapted to this proof, it is given below.

$$\left(\frac{\partial P}{\partial T}\right)_\rho \left(\frac{\partial T}{\partial \rho}\right)_P \left(\frac{\partial \rho}{\partial P}\right)_T = -1$$

Rearranging this relation gives the following equation.

$$\left(\frac{\partial T}{\partial \rho}\right)_P = -\frac{1}{\left(\frac{\partial P}{\partial T}\right)_\rho \left(\frac{\partial \rho}{\partial P}\right)_T}$$

Again, we must utilize an identity we learned in class: the reciprocity relationship. It is given below, and the relevant correlation is also shown.

$$\left(\frac{\partial x}{\partial y}\right)_z = \frac{1}{\left(\frac{\partial y}{\partial x}\right)_z}$$

$$\left(\frac{\partial \rho}{\partial T}\right)_P = \frac{1}{\left(\frac{\partial T}{\partial \rho}\right)_P}$$

Combining the relationships we get from the cyclic and reciprocity identities gives the following equation.

$$\left(\frac{\partial \rho}{\partial T}\right)_P = -\left(\frac{\partial \rho}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho$$

Now, we can input this into the bracketed term in equation 6. By doing so, the bracketed term becomes equal to zero and cancels out.

$$-\left(\frac{\partial P}{\partial T}\right)_\rho \left(\frac{\partial \rho}{\partial P}\right)_T - \left[-\left(\frac{\partial \rho}{\partial P}\right)_T \left(\frac{\partial P}{\partial T}\right)_\rho \right] = 0$$

$$\frac{T}{\rho^2} \left[-\left(\frac{\partial P}{\partial T}\right)_\rho \left(\frac{\partial \rho}{\partial P}\right)_T - \left(\frac{\partial \rho}{\partial T}\right)_P \right] = 0$$

As a result, we get the following equation for the derivative of the right side.

$$\left(\frac{\partial Y}{\partial P}\right)_T = V - T \left(\frac{\partial V}{\partial T}\right)_P$$

Now, recognize that this is exactly equal to equation 2. Therefore, we have proven $\left(\frac{\partial X}{\partial P}\right)_T = \left(\frac{\partial Y}{\partial P}\right)_T$. From this, we can write the following equation.

$$\frac{\partial}{\partial P} [X(P, T) - Y(P, T)]_T = 0$$

Since we have proven that the difference between the derivatives of the two sides of the initial equation with respect to pressure at constant temperature is equal to zero, we can conclude that this difference either relies solely on a single variable temperature, or that the difference is always zero and the two sides are equal. So, the next step to confirm this identity is to prove that both sides are equal to each other at any given temperature. To start, let's evaluate both sides as pressure approaches zero, or in other words, as an ideal gas.

Take the following equation and set integration boundary P equal to zero.

$$X(P, T) = \int_0^P (1 - \beta T) V dP$$

The integration boundary becomes zero, so it is evaluated to be zero. We can state the following.

$$X(0, T) = 0$$

Now let's evaluate the other side at zero.

$$Y(P, T) = RT(Z - 1) - \int_0^P RT^2 \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho}$$

For an ideal gas, Z is equal to 1. So, the left term cancels to become zero. Additionally, the value for Z will not change for an ideal gas, let alone with respect to temperature. So, the integrand term evaluates to zero. Therefore, the following is true

$$Y(0, T) = 0$$

To summarize, we have proven that differentials of both sides of the identity with respect to pressure at a constant temperature are equal. Additionally, we have evaluated both sides of the identity to be equal at different temperatures under ideal gas conditions.

In other words, we proved both sides vary equally according to pressure, and they have equal constants of integration. Therefore, the equality of the identity holds true for all values of pressure and temperature. Thus, the identity is true.

$$\int_0^P (1 - \beta T) V dP = RT(Z - 1) - \int_0^P RT^2 \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho} \quad \text{where} \quad Z = \frac{PV}{RT}$$

$$X(P, T) = \int_0^P (1 - \beta_T T) V dP$$

$$Y(P, T) = RT(Z - 1) - \int_0^P RT^2 \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho}$$

$$\left(\frac{\partial X}{\partial P} \right)_T = \left(\frac{\partial Y}{\partial P} \right)_T$$

$$X(0, T) = 0 \qquad Y(0, T) = 0$$

Problem B

We want to calculate the fugacity of dry ice at the given pressure and temperature. Before making any calculations, let's list out some assumptions and quick pieces of information we can get from the problem statement. Note, we used data from engineeringtoolbox.com to get our numbers for calculations. It is referenced properly at the end of this report.

- Assume CO₂ follows the Peng-Robinson EOS
- Heat of fusion is 9.02 kJ/mol
- Approximate melting point is 56.6°C

The plan to calculate the fugacity of dry ice is to start by calculating the fugacity at the melting point of dry ice at the given pressure. We will do this by using the Zcomp and PhiC functions in the CHE calculator. Then, once we have the fugacity at the melting point, we will use equation 4.158 from the textbook to calculate the solid state fugacity.

First, we listed the necessary known values, and added CO₂ as a component in the CHE calculator. Then we used the Zcomp function to get the compressibility factor. The window is shown below.

Function Arguments

ZComp

MolFrac	comp1	=	"CO2"
T	D4	=	56.6
P	D5	=	101.325
Zo	0	=	0
Maxite		=	

Calculates the Compressibility Factor.

Maxite Optional: Maximum Number of Iterations [100].

Formula result = 0.504620462

[Help on this function](#) OK Cancel

Next, using the calculated compressibility factor, we used the PhiC function to calculate the fugacity coefficient. The window is shown below.

Function Arguments

PhiC

MolFrac	comp1	=	"CO2"
CompCol	comp1	=	"CO2"
T	D3	=	-120
P	D5	=	101.325
Z	D7	=	0.504620462

Calculates the Fugacity Coefficient.

Z Compressibility Factor.

Formula result = 0.009862716

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Now, we multiply the fugacity coefficient by the given pressure to get a value of 0.99934 bar. We obtain the following table.

Parameter	Value
Temperature, C	-120
Melting Point, C	56.6
Pressure, bar	101.325
Compressibility	0.50462
Fugacity constant	0.00986
Liquid fugacity, bar	0.99934

Now, we are ready to use equation 4.158. It is shown below.

$$f_s = f_L \exp \left[\frac{\Delta H^{\text{fus}}}{RT} \left(1 - \frac{T}{T_m} \right) \right]$$

To cleanly execute this, we expanded our above table to include the necessary values in their corresponding units. The expanded table is shown below, with the final row containing the calculated value for the fugacity of dry ice.

Parameter	Value
Temperature, C	-120
Melting Point, C	56.6
Pressure, bar	101.325
Compressibility	0.50462
Fugacity constant	0.00986
Liquid fugacity, bar	0.99934
R, J/mol*K	8.314
Heat of fusion, J/mol	9020
Temperature, K	153.15
Melting Point, K	329.75
Solid fugacity, bar	44.3998

Our final answer for problem B is shown below.

$$f_s = 44.3998 \text{ bar}$$

Problem C

For this problem, we want to calculate the mass fraction of dry ice formed after expansion through a throttling valve. Firstly, let us assume that the process in the throttling valve is adiabatic and isenthalpic. In other words, the outlet and inlet enthalpies are equivalent. Given the inlet pressure, temperature, and Peng-Robinson EOS, we can calculate this enthalpy.

First, let's determine the state of the inlet. It will help to know the saturated pressure at the given temperature. To obtain that value, we can simply use the EOSPsat function in the calculator.

Function Arguments

EOSPsat

CompCol	comp1	=	"CO2"
T	-35	=	-35
P0		=	
Maxite		=	
Tol		=	

= 11.8738596

Calculates the Saturated Vapor Pressure [bar] with the Selected EOS.

P0 Optional: Initial Guess for Saturated Vapor Pressure [bar].

Formula result = 11.8738596

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$$P_{sat} = 11.874 \text{ bar}$$

The given pressure is 60 atm, so even without conversions, it is easy to see that it is greater than the saturated pressure. This indicates that the inlet pressure is subcooled liquid. Knowing that it is purely subcooled liquid makes calculations a little easier.

Now, let's calculate the inlet compressibility factor by using ZComp again. We make an initial guess of zero, since we are dealing with a liquid. The window is shown below.

Function Arguments

ZComp

MolFrac	comp1	=	"CO2"
T	-35	=	-35
P	60.795	=	60.795
Zo	0	=	0
Maxite		=	

= 0.116953112

Calculates the Compressibility Factor.

Maxite Optional: Maximum Number of Iterations [100].

Formula result = 0.116953112

[Help on this function](#) OK Cancel

To calculate the inlet enthalpy, we can use the Enthalpy function. The window is shown below.

Function Arguments

Enthalpy

MolFrac	comp1	=	"CO2"
T	I3	=	-35
P	I5	=	60.795
Z	I7	=	0.116953112
Hfo		=	

Calculates the Molar Enthalpy [kJ/kmol].

Hfo Optional: Enthalpy of Formation [kcal/mol].

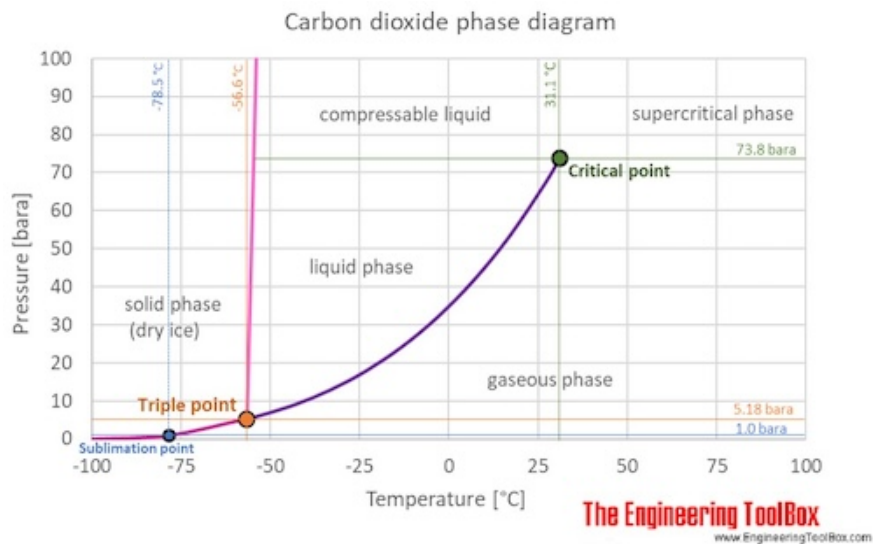
Formula result = -410599.2858

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The calculated values for the inlet stream are summarized in the table below. The enthalpy is highlighted yellow because it should be kept constant after the process.

Parameter	Inlet
Temperature, C	-35
Pressure, bar	60.795
Compressibility	0.11695
Liquid enthalpy, kJ/kmol	-410599

Based on the graph below, we can surmise that the exit stream consists of vapor and dry ice. We know it is a mixture containing dry ice from the problem statement, and the pressure of 1 atm lines up with the solid-vapor line on the graph.



Now, we can perform similar calculations as before to get a cell that calculates the enthalpy of the outlet stream. The only difference is that we need separate cells to hold the values for the solid state enthalpy and the vapor state enthalpy, which we can later combine via the lever rule.

To get different values of enthalpy for the solid state and vapor state, we obtain different compressibility factors. We get these different values by using different starting guesses. For the solid, we guess zero, and

for the vapor, we guess one. The windows for those functions are shown below.

Function Arguments

ZComp

MolFrac comp1 = "CO2"

T M3 = -88.22294359

P M4 = 1.01325

Zo 1 = 1

Maxite = 0.978677199

Calculates the Compressibility Factor.

Maxite Optional: Maximum Number of Iterations [100].

Formula result = 0.978677199

[Help on this function](#) OK Cancel

Function Arguments

ZComp

MolFrac comp1 = "CO2"

T M3 = -88.22294359

P M4 = 1.01325

Zo 0 = 0

Maxite = 0.002162222

Calculates the Compressibility Factor.

Maxite Optional: Maximum Number of Iterations [100].

Formula result = 0.002162222

[Help on this function](#) OK Cancel

Then, we use the same enthalpy functions as before to get a value for vapor and solid enthalpies. It is the exact same as the functions from before, so we will omit the windows here. We combine these via the lever rule, as depicted below.

$$h = (1 - x)h_v + xh_s$$

Finally, we set up a cell to take the difference between the inlet and outlet enthalpies. We use goal seek to make this cell equal to zero by changing the vapor fraction. All calculated values are shown in the following table.

Parameter	Inlet	Outlet
Temperature, C	-35	-88.2229
Pressure, bar	60.795	1.01325
Compressibility	0.11695	
Liquid enthalpy, kJ/kmol	-410599	-410599
Vapor enthalpy, kJ/kmol		-397756
Solid enthalpy, kJ/kmol		-414736
Vapor fraction		0.2436
Compressibility Vapor		0.97868
Compressibility Solid		0.00216

Everything is in a molar basis, however there is no need to convert anything to a kilogram basis since the vapor fraction would come out to the same value due to there being no other molar masses for other substances involved. The final step is to subtract the vapor fraction from 1, yielding the final answer below.

$$1 - 0.2436 = 0.7564$$

$$\text{Mass fraction of dry ice} = 0.7564$$